Электрическое поле

① min. заряд $1{,}6 \cdot 10^{-19}$ Кл

Закон сохр. заряда

② Напряженность эл. поля

$\vec{E}$ $\quad \vec{F} = q\vec{E}$

③ Закон Кулона

СИ: $F = k\frac{q_1 q_2}{r^2}$ $\quad k = 9 \cdot 10^9$ $\quad [q] = Кл$

СГСЭ: $F = \frac{q_1 q_2}{r^2}$ $\quad [q] = \sqrt{дин} \cdot см$

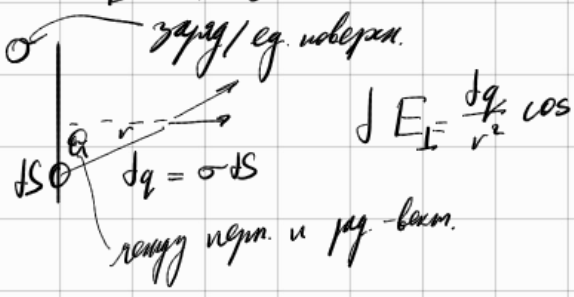
④ $E = \frac{q}{r^2}$ (СГСЭ)

$\vec{E} = \frac{q}{r^2}\vec{e}_r$ $\quad \vec{e}_r = \frac{\vec{r}}{r}$

$\vec{E} = \frac{q}{r^3}\vec{r}$

⑤ Принцип суперпозиции

$\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots$



$dE_\perp = \frac{dq}{r^2}\cos\theta = \frac{\sigma dS \cos\theta}{r^2}$



$d\Omega = \frac{dS\cos\theta}{r^2}$ — телесн. угол

$dS_\perp$

$E_\perp = \sigma\Omega$

сумму: фигура



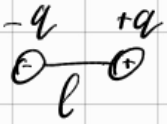
$\Omega = 2\pi(1 - \cos\theta)$

Электрический диполь



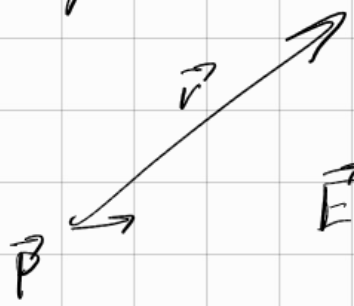
$$\vec{P} = \sum_{i=1}^{N} q_i \vec{r}_i$$

$$Q = \sum_{i=1}^{N} q_i = 0$$



– эл. диполь $\quad \vec{P} = q\vec{l}$

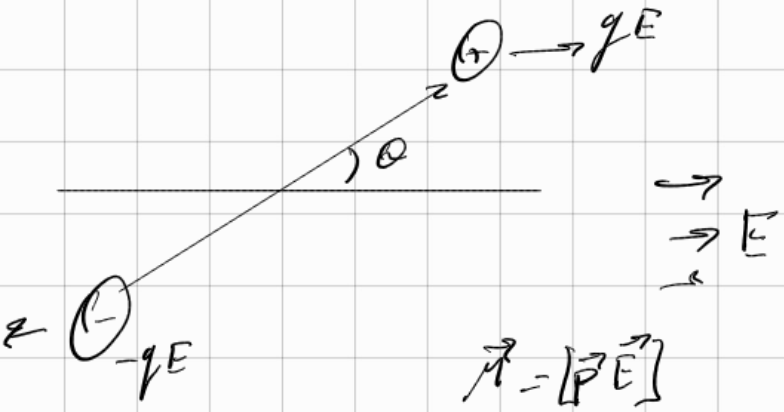
Эл. поле диполя



$$\vec{E}(\vec{r}) = \frac{3(\vec{P}\cdot\vec{r})\vec{r}}{r^5} - \frac{\vec{P}}{r^3}$$

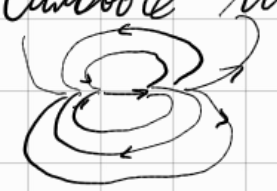
Момент сил

$$M = pE\sin\theta$$



$$\vec{M} = [\vec{P}\vec{E}]$$

Силовые линии

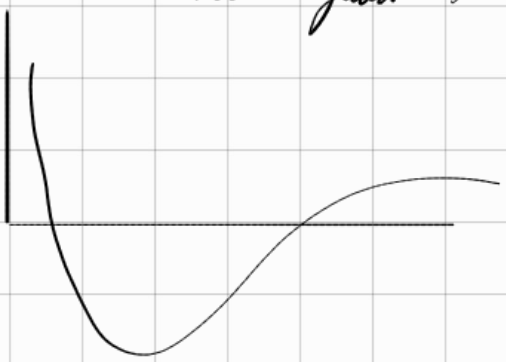


Сила вз-я 2 диполей $\quad F = P\frac{dE}{dr}$

$$\vec{F} = -q\vec{E}(r) + q\vec{E}(r+l)$$

$F = P\frac{dE}{dr}$ $\quad E = \frac{2P_1}{r^3}$ $\quad F = -\frac{6P_1P_2}{r^4}$

Потенциал Леннард - Джонса



$U = \frac{Q_1}{r^{12}} - \frac{Q_2}{r^6}$ $\quad F = -\frac{dU}{dr} \sim \frac{1}{r^7}$

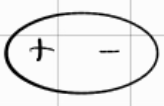
реальность

пояснение к

1) Индукционное вз-е

$P_1 \neq 0$ $\quad P_2 = 0$

$P_2 = \alpha E = \frac{2\alpha P_1}{r^3}$



$P = \alpha E$

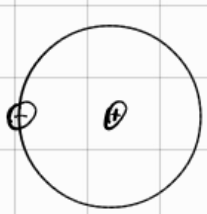
, если подставить в $F = \frac{6P_1P_2}{r^4}$, то $\sim \frac{1}{r^7}$

2) Ориентационные силы

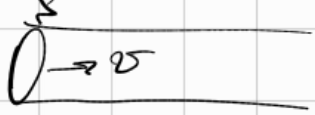
$P_1 \neq 0$ $\quad P_2 \neq 0$ $\quad P_2 \langle \cos\theta \rangle \sim \frac{P_2}{r^3}$

3) Дисперсионное

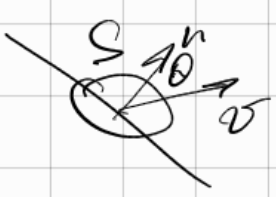
$P_1 = 0$ $\quad P_2 = 0$



Теорема Гаусса



$\Phi = S \cdot v$



$\Phi = v \cdot S \cdot \cos\theta = \vec{v}\vec{S}$

$\vec{S} = S \cdot \vec{n}$ $\quad \Phi = \int_S \vec{E}\,d\vec{S} = \int_S E_n\,dS$



(СГС) $\oint \vec{E}\,d\vec{S} = 4\pi Q$

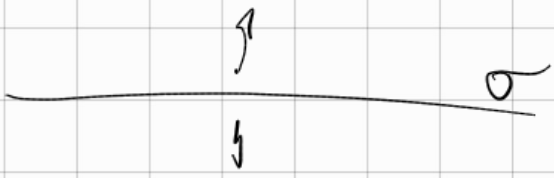
$\Phi = \oint \vec{E}\,d\vec{S} = \frac{Q}{\varepsilon_0}$ (СИ)



$E = \frac{q}{r^2}$

$\Phi = E 4\pi r^2 \Rightarrow \Phi = 4\pi q$

Поле плоскости



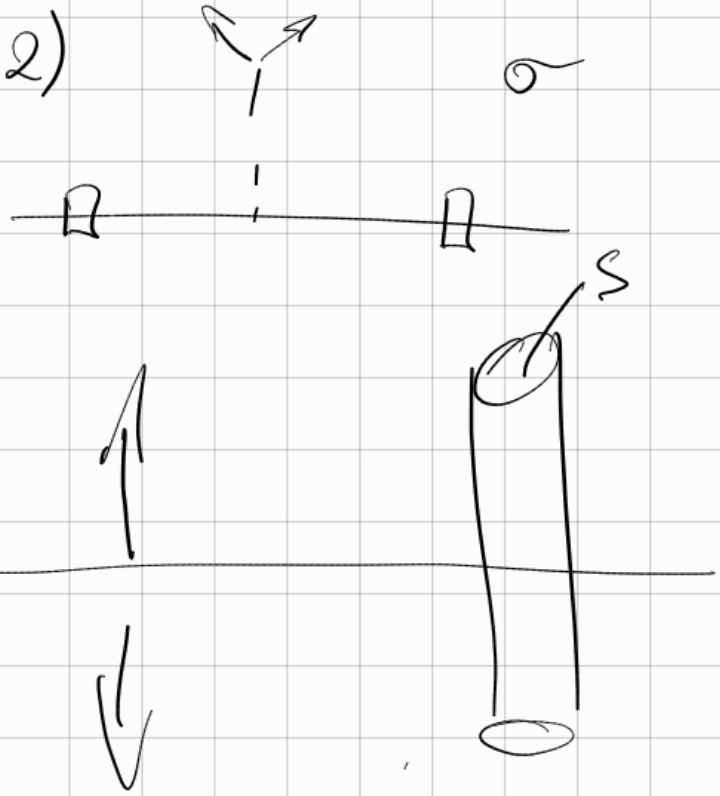
$E = \sigma\lambda = 2\pi\sigma$

$\oint \vec{E} d\vec{S} = 4\pi Q$ СГС $\quad E \cdot 4\pi R^2 = 4\pi Q \Rightarrow E = \frac{Q}{R^2}$

1)



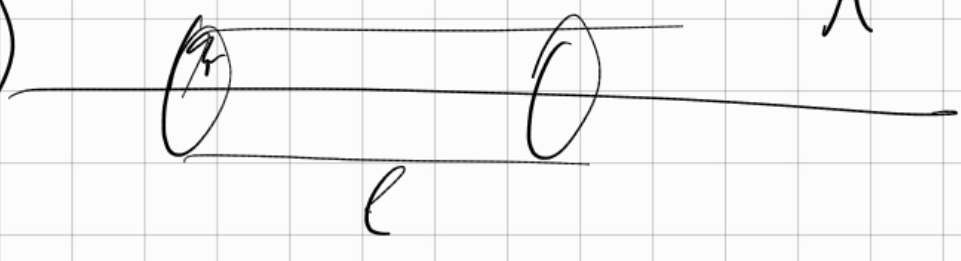
2)



$\Phi = 2ES = 4\pi\sigma S$

$\Rightarrow E = 2\pi\sigma$

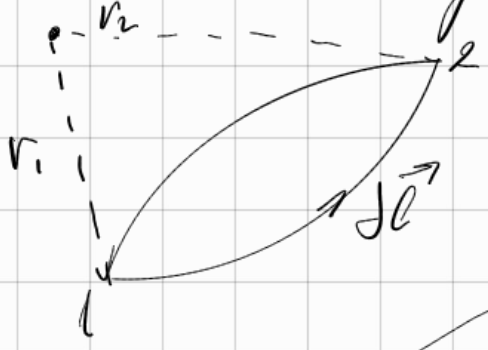
3)



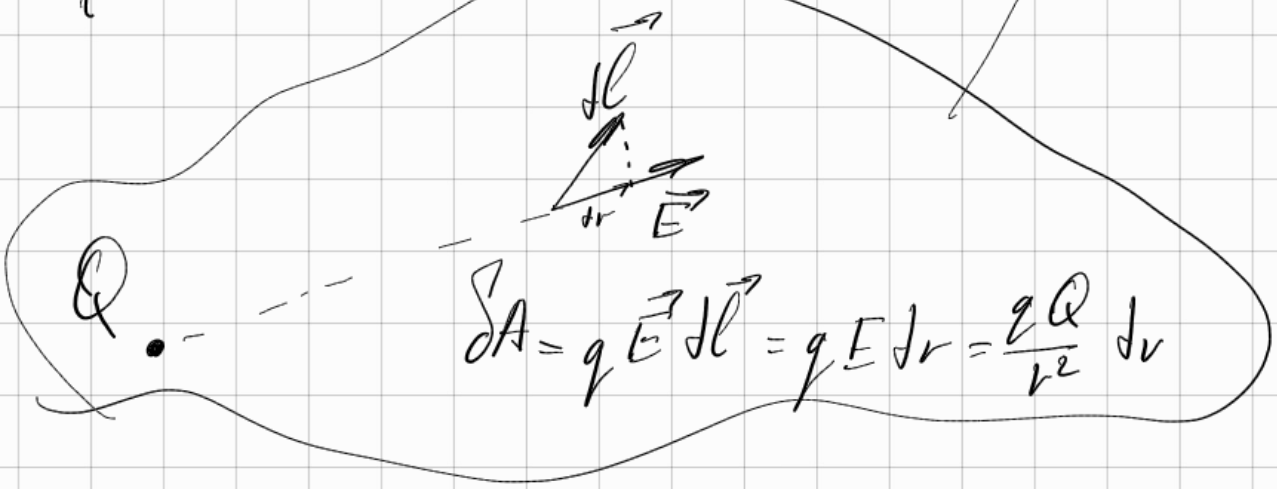
$\Phi = 2\pi r \ell E = 4\pi\lambda\ell$

$\Rightarrow E = 2\lambda / r$

Потенциал электро-статического поля

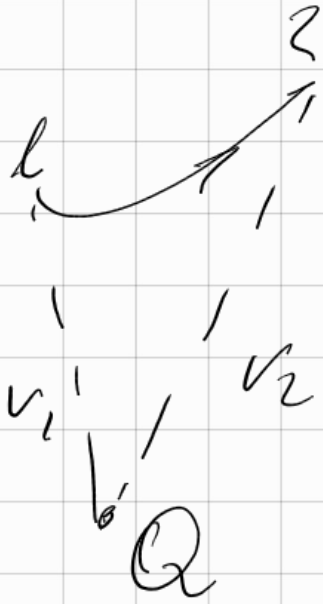


$$A = \int_1^2 q\vec{E}\,d\vec{l} \qquad A = qQ\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$



$$\delta A = q\vec{E}\,d\vec{l} = qE\,dr = \frac{qQ}{r^2}\,dr$$

Следствие $\oint \vec{E}\,d\vec{l} = 0$



$$\Pi_2 - \Pi_1 = \int_1^2 \vec{F}_{внеш} \cdot d\vec{l}$$

$$\vec{F}_{внеш} = -q\vec{E}$$

$$\Pi_2 - \Pi_1 = -q\int_1^2 \vec{E}\,d\vec{l}$$

$$\varphi = \frac{\Pi}{q}$$

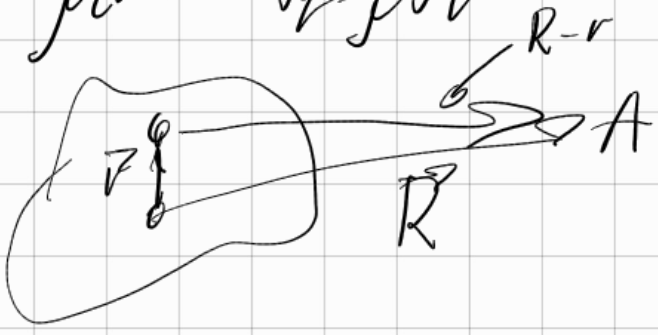
Потенциал

$$\varphi_2 - \varphi_1 = -\int_1^2 \vec{E}\,d\vec{l}$$

Принцип суперпозиции: $\varphi = \varphi_1 + \varphi_2 + \varphi_3$

1) $\varphi = \frac{Q}{r}$

2) $\rho(\vec{r})$ $dq = \rho dV$



РАСПР. ЗАР. В ПРОСТР.

$$\varphi = \int_V \frac{\rho(\omega) dV}{|\vec{R} - \vec{r}|}$$

3) Пот. диполя

$$\varphi = \frac{\vec{p}\vec{r}}{r^3}$$

Энергия диполя во внешн. поле:

$U = -\vec{p}\vec{E}_{внеш}$ $\vec{p} = const$

$U = -\frac{1}{2}\vec{p}\vec{E}_{внеш}$ $p \propto E_{внеш}$

$d\varphi = -\vec{E}\,d\vec{l}$ $-(E_x dx + E_y dy + E_z dz)$ $d\vec{l} = (dx; dy; dz)$

$f(x) \Rightarrow df = \frac{df}{dx} dx$

$\frac{\partial f}{\partial x}; \frac{\partial f}{\partial y}; \frac{\partial f}{\partial z}$

$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$

$E_x = -\frac{\partial \varphi}{\partial x}$

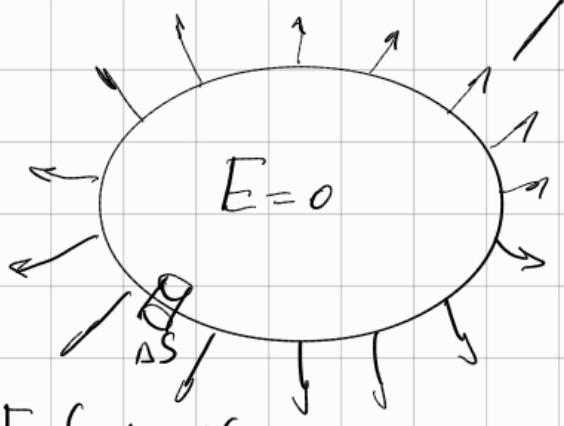
$E_y = -\frac{\partial \varphi}{\partial y}$

$E_z = -\frac{\partial \varphi}{\partial z}$

$\vec{E} = -grad\,\varphi = -\vec{\nabla}\varphi$

$\vec{\nabla} = \left(\frac{\partial}{\partial x}; \frac{\partial}{\partial y}; \frac{\partial}{\partial z}\right)$

Электрическое поле в присутствии проводников



$j = \sigma \vec{E}$

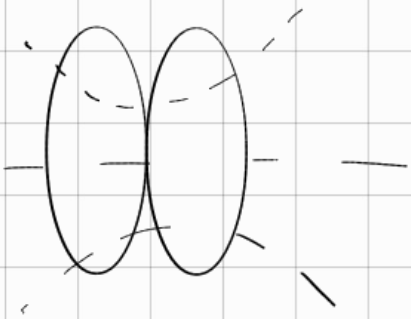
Эквипотенциальная пов-ть $\varphi = const$

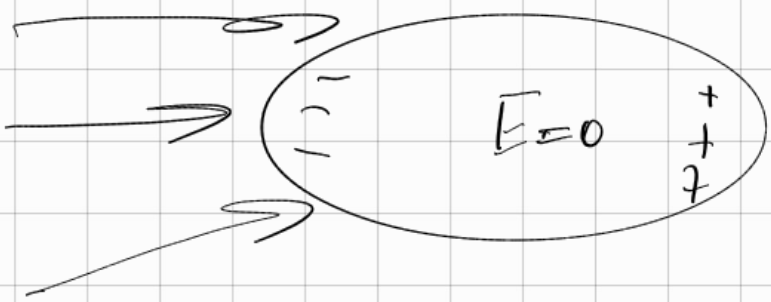
$E \Delta S = \pi \sigma \Delta S$

$E = \pi \sigma$ СГСЭ

$E = \frac{\sigma}{\varepsilon_0}$ СИ

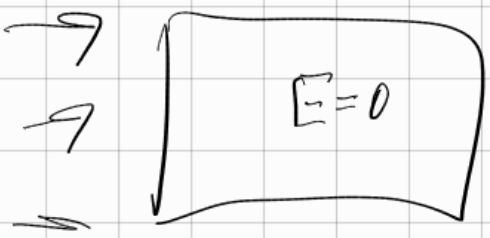
$\varphi_2 - \varphi_1 = -\int_1^2 \vec{E} d\vec{l}$





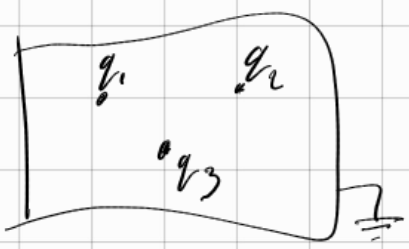
Поз-ть эквипотенц., а поле внутри равно нулю

Клетка Фарадея



Замкн. метал. пов-ть во внешнем поле внутри поле = 0

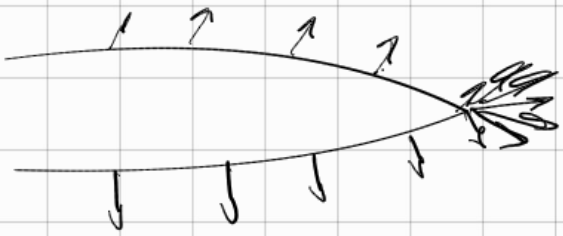
Заземление



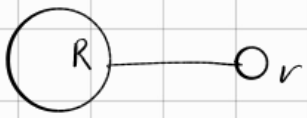
$\varphi = 0$

$E = 0$

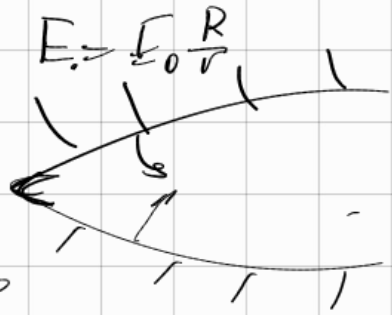
Д. шар в поле бегущей волны



$$\varphi = \frac{Q}{R} = \frac{q}{r}$$

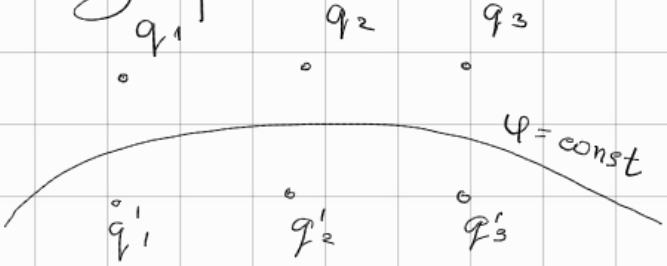


$$E_0 = \frac{q}{r^2} \quad E_0 = \frac{Q}{R^2}$$

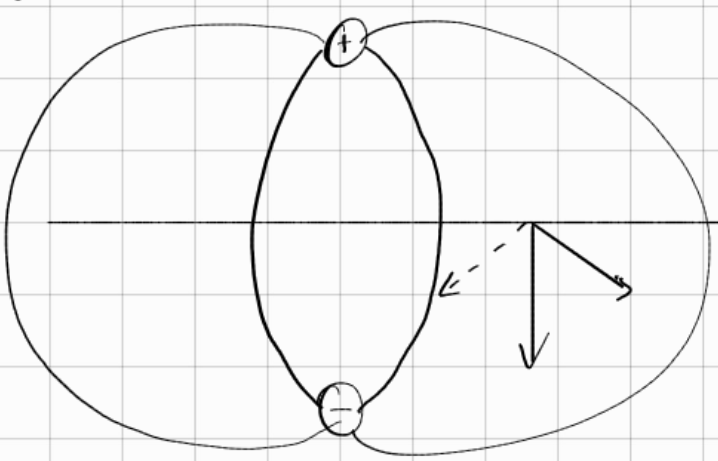


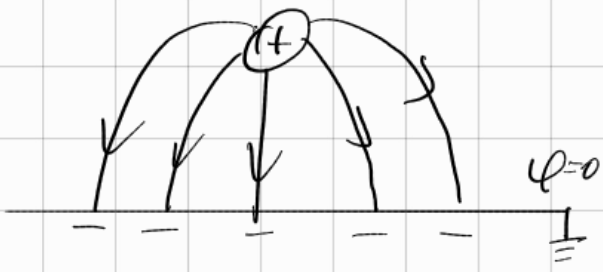
$$E_0 = E_0 \frac{R}{r}$$

Метод изображений

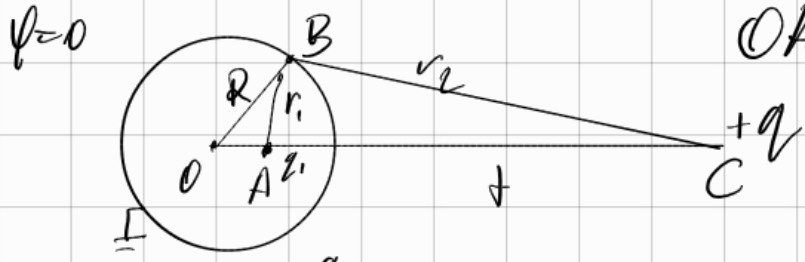


08.09.26





2. Заряд + шар. сфера

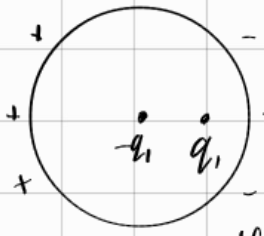


$$Q_{сф} = q_1 \quad \varphi = \frac{q}{r_2} + \frac{q_1}{r_1} = 0 \quad q_1 = q\frac{r_1}{r_2} = -q\frac{R}{d}$$

$\triangle OBA \sim \triangle OBC$

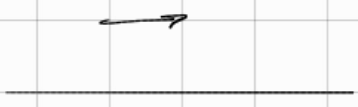
$K = \frac{R}{d} = \frac{a}{R} \Rightarrow a = \frac{R^2}{d} \quad q_1 = q\frac{R}{d}$

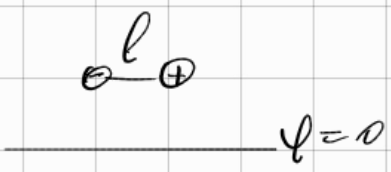
$F = \frac{qq_1}{(d-a)^2}$



$Q_{сф} = 0 \qquad \varphi = -\frac{q_1}{R} = \frac{q}{d}$

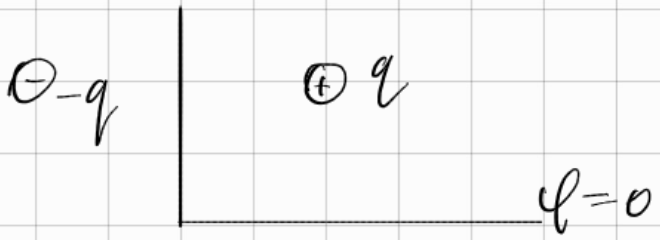


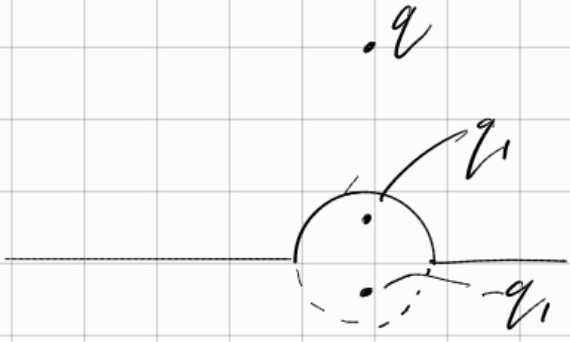




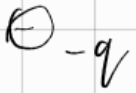






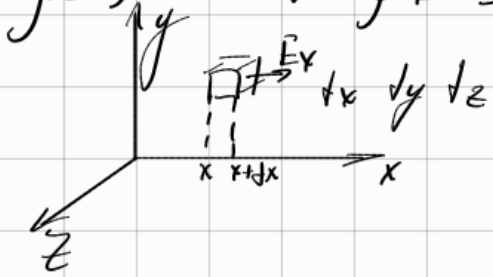






Теорема Гаусса в дифф. форме

$\rho(\vec{r})$ $\qquad \Phi = \oint \vec{E}\, d\vec{S} = 4\pi Q$ СГСЭ



$\vec{E} = (E_x; E_y; E_z)$

$d\varphi_x = E_x(x+dx; y; z)\, dy\, dz - E_x(x; y; z)\, dy\, dz$

$d\varphi_x = \frac{\partial E_x}{\partial x} dx\, dy\, dz = dV$

$d\varphi_y = \frac{\partial E_y}{\partial y} dV \qquad d\varphi_z = \frac{\partial E_z}{\partial z} dV$

$d\varphi = d\varphi_x + d\varphi_y + d\varphi_z = 4\pi\rho\, dV$

$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 4\pi\rho$$

$$\vec{\nabla}\vec{E} = 4\pi\rho$$

дивергенция E

$$\operatorname{div}\vec{E} = 4\pi\rho$$

Уравнение Пуассона

$$\vec{\nabla}\vec{E} = 4\pi\rho \qquad \vec{E} = -\vec{\nabla}\varphi$$

$$\vec{\nabla}^2\varphi = -4\pi\rho$$

$$\vec{\nabla}^2 = \Delta \qquad \Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\Delta\varphi = -4\pi\rho$$

$$\frac{\partial^2\varphi}{\partial x^2} + \frac{\partial^2\varphi}{\partial y^2} + \frac{\partial^2\varphi}{\partial z^2} = -4\pi\rho$$

$\varphi = const$

$\rho = 0$ - У-е Лапласа

$\Delta\varphi = 0$

Переписать у Алеева

15.09

$\vec{\nabla}\vec{D} = 4\pi\rho_{своб} = 0$

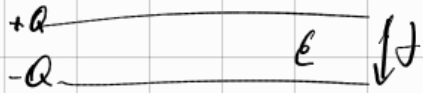
$\vec{D} = \varepsilon\vec{E}$

$\varepsilon\,\vec{\nabla}\vec{E} = 0$

$\vec{\nabla}\vec{E} = 4\pi\rho_{пол} = 0 \quad \varepsilon = const$

Энергия электростат. поля

$U = \varphi_2 - \varphi_1 \quad Q = CU$



$D = 4\pi\sigma$

$E = \frac{4\pi\sigma}{\varepsilon} \Rightarrow \frac{U}{d} = \frac{4\pi Q}{\varepsilon S} \Rightarrow$

$\Rightarrow Q = \frac{\varepsilon S}{4\pi d} U$

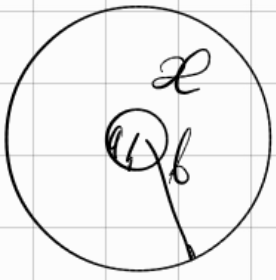
$C = \frac{\varepsilon S}{4\pi d}$ СГСЭ

$C = \frac{\varepsilon\varepsilon_0 S}{d}$ СИ

Изолированный электрод $Q = C\varphi$

$\varphi = \frac{Q}{R} \Rightarrow C = R$





$D = \frac{2\alpha}{r}$

$E = \frac{2\alpha}{\varepsilon r} \quad U = \int_a^b E\,dr =$

$= 2\frac{Q}{\ell}\ln\frac{b}{a}$

$C = \frac{\varepsilon\ell}{2\ln b/a}$

Энергия электростат. поля



$dW = SA = U\,dQ = u\,C\,du$

$W = \frac{Cu^2}{2} \quad W = \frac{Q^2}{2C} \quad W = \frac{Qu}{2}$

① $W = \frac{1}{2}\sum_{i=1}^{N} Q_i\varphi_i$ – энергия электростат. поля

$Q_i \to \rho\,dV$

② $W = \frac{1}{2}\int_V \rho\varphi\,dV$

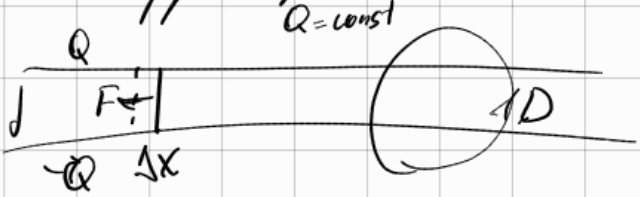
Плотность энергии

$\vec{D} = \varepsilon \vec{E}$

$$w = \frac{\vec{E}\vec{D}}{8\pi} = \frac{\varepsilon E^2}{8\pi} = \frac{D^2}{8\pi\varepsilon}$$

$$W = \int_U w\,dU$$

Пондеромоторные силы



$F_{\Delta x} = -\Delta W \quad W = \frac{1}{2}\frac{Q^2}{C}$

$F = -\left(\frac{dW}{dx}\right)_Q$

$F = -\frac{Q^2}{2}\frac{d(1/C)}{dx} = \frac{Q^2}{2C^2}\frac{dC}{dx} =$

$= \frac{U}{2}\frac{dC}{dx}$

Энергетический метод вычисления сил (виртуальных перемещений)

$\Delta W = \frac{U \Delta Q}{2}$

$F_{\Delta x} = -\Delta W + \Delta A = \frac{1}{2} U \Delta Q = \Delta W$

$\Delta A = U \Delta Q$

$F = \left(\frac{dW}{dx}\right)_U \quad W = \frac{CU^2}{2}$

$F = \frac{U^2 dC}{2\,dx}$

$\Delta W = \Delta U \frac{\varepsilon E^2}{8\pi} - \Delta U \frac{E^2}{8\pi} = \frac{E^2}{8\pi}(\varepsilon - 1)\,a\,d\,\Delta x$

$F = \frac{\varepsilon E^2}{8\pi} - \frac{E^2}{8\pi} a d = (w_2 - w_1) \underbrace{a d}_{S_T}$

$F = \left(\frac{dW}{dx}\right)_U$

$P = \frac{F}{S_T} = w_2 - w_1$

$P = w_2 - w_1$

Сила, действующая на единицу площади поверхности раздела диэлектриков с различными ε

A) 